Xiaodong Huang

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EDUCATION

University of Minnesota Twin Cities (UMN) Minneapolis, MN
Bachelor of Science in Statistics; Minors in Computer Science and Mathematics *Sep. 2023 – May 2026*

- **GPA**: 3.92/4.00

RESEARCH INTERESTS

Trustworthy clinical AI for translating EHR-based prediction into actionable longitudinal decision support, with interests in real-world evidence, causal inference, and knowledge- and graph-guided multi-agent systems.

PUBLICATIONS & MANUSCRIPTS

- Jin, Y., **Huang, X.**, Wang, Y., Wacker, D. A., Puskarich, M., and Xie, F. (2025). *Development and Multicenter External Validation of a Data-Driven Score for Early and Rapid Identification of Sepsis in Emergency Departments*. Under revision at *eClinicalMedicine*. Preprint available on *medRxiv*. doi:10.1101/2025.09.27.25336784
- **Huang, X.**, Patel, S., Jin, Y., Mai, X., and Xie, F. (2026). *ActionRef: Learning Reference Thresholds and Guiding Clinical Risk Threshold Selection*. Manuscript in preparation.

CONFERENCE & WORKSHOP PAPERS

- Jin, Y.*, **Huang, X.***, Chandu, H. P., Xiong, X., Lin, M., and Xie, F. (2026). *RADAR: A Guideline-Grounded Multi-Agent System for the Emergency-Department Sepsis Pathway*. Accepted to the *NeurIPS 2026 Workshop on GenAI4Health*, Demonstration Paper Track. *Co-first authors.

RESEARCH EXPERIENCE

University of Minnesota Twin Cities *May 2025 – Present*
Supervisor: **Dr. Feng Xie**, Assistant Professor, Division of Computational Health Sciences
Clinical Machine Learning and Decision Systems for Emergency Care

- **Research focus:** Translate EHR-based risk estimates into interpretable and operationally feasible decision support across heterogeneous emergency departments.
- **Sepsis risk prediction:** Contributed to the development and multicenter external validation of the Emergency Sepsis Risk Predictor using more than 2.2 million ED encounters across UMN, BIDMC/MIMIC-IV-ED, and Stanford MC-MED; harmonized cohort definitions and developed reproducible Python and R evaluation pipelines, with AUROCs ranging from 0.820 to 0.892 across the three cohorts.
- **Decision-threshold learning:** Led a behavior- and capacity-aware framework that converts continuous risk scores into clinically interpretable Low/Mid/High decision tiers; demonstrated stable operating characteristics across 500 patient-level bootstrap resamples and retained strict threshold ordering in all 100 re-estimation replicates in both MIMIC-IV-ED and MC-MED.
- **Sepsis phenotyping:** Developed representation-learning pipelines to characterize clinically interpretable sepsis subgroups and evaluate phenotype stability, outcome separation, and transportability across independent EHR cohorts.
- **Longitudinal decision support:** Contributed to RADAR, a guideline-grounded and graph-guided multi-agent system that maintains a shared patient state across recognition, treatment, safety, and disposition stages in emergency sepsis care.

Peking University *May 2026 – Present*
Research collaboration with **Dr. Shenda Hong**, Assistant Professor, National Institute of Health Data Science
EHR-Based Target Trial Emulation of Antibiotic Timing Strategies in Emergency Department Sepsis

- **Research focus:** Estimate how antibiotic timing affects sepsis outcomes and whether effects differ across patient phenotypes while addressing time-related bias in observational EHR studies.
- Develop a target trial emulation with explicit eligibility criteria, ED arrival as time zero, aligned treatment and follow-up windows, and prespecified early-versus-delayed antibiotic strategies.
- Integrate a clinically derived sepsis phenotype classifier as a prespecified effect modifier to evaluate heterogeneous treatment effects, with attention to immortal-time bias, treatment selection, and dynamic confounding.

RESEARCH EXPERIENCE (CONTINUED)

University of Minnesota Twin Cities *Nov. 2024 – Apr. 2025*
Supervisor: **Dr. Jie Ding**, Associate Professor, School of Statistics
AI and Data Science for Collaborative Research and Life Sciences

- **Research focus:** Investigated how LLM-based agents can reduce friction in collaborative scientific workflows and extract reusable knowledge from biomedical documents.
- Contributed to an AI-agent platform for collaborative research, including document understanding, LaTeX diagnostics, visualization, file workflows, and interactive research automation.
- Supported LLM-based biomedical text mining and structured information extraction for next-generation scientific research assistants.

PRESENTATIONS

- Jin, Y., **Huang, X.**, Wang, Y., Wacker, D. A., Puskarich, M., and Xie, F. (2025). *Development and Multicenter External Validation of a Data-Driven Scoring System for Early and Rapid Identification of Sepsis in Emergency Departments*. Oral presentation at the **AMIA 2025 Annual Symposium**.
- **Huang, X.**, Patel, S., Jin, Y., Mai, X., and Xie, F. (2026). *From Risk Prediction to Clinical Action: A Universal Behavior-Informed Framework for Learning and Governing Clinical Decision Thresholds from Real-World Practice*. Oral presentation at the University of Minnesota Undergraduate Research Symposium.

HONORS & AWARDS

- **2026 Love of Learning Grant**, The Phi Beta Kappa Society — Jun. 2026
- **Elected Member**, The Phi Beta Kappa Society — May 2026
- **UROP Research Award (competitive)**, University of Minnesota (Spring 2026) — Dec. 2025
- **Dean's List**, University of Minnesota — 2023–2026

ADDITIONAL PROFESSIONAL EXPERIENCE

Hubei Hanjiang Guotou Digital Industry Group Co., Ltd. *Jun. 2025 – Dec. 2025*
Data Analyst Intern

- Built SQL/Python data pipelines that cleaned and standardized more than 60,000 IoT and municipal records for a city-management platform and supported LLM-assisted retrieval over operational data.

Shenzhen Manjinghua Investment Group *May 2024 – Aug. 2024*
R&D Intern, Big Data R&D Center

- Analyzed multi-source operational data and developed regression-based foot-traffic forecasts and dashboards in Python and MATLAB to support resource allocation and predictive-maintenance analyses.

METHODS & TECHNICAL SKILLS

Programming	Python, R, SQL, Stata, Java, MATLAB
Libraries	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, pandas, NumPy
Tools	Git, LaTeX
Clinical/Statistical	EHR cohort design, clinical prediction modeling, decision-threshold design, external validation, calibration, bootstrap resampling, survival analysis, target trial emulation
AI Methods	Representation learning, deep clustering, retrieval-augmented generation, multi-agent systems

SELECTED COURSEWORK

Statistical Theory I & II; Regression and Correlated Data; Statistical Machine Learning I & II; Applied Multivariate Methods; Nonparametric Methods; Data Structures and Algorithms; Computational Linear Algebra.